Setup

1. **Install Docker**

```
sudo apt update
sudo apt install -y docker.io
sudo systemctl start docker
sudo systemctl enable docker
```

2. **Create new docker user group**

```
sudo usermod -aG docker $USER
newgrp docker
```

3. **Restart the instance**

Can just stop and restart VM

4. **Get ONOS**

```
git clone https://gerrit.onosproject.org/onos
git checkout 2.2.1
```

5. **Generate Atomix configuration files using onos (don't need to do this if you already have a "atomix-configs" folder containing 3 "atomix-[num]-conf" folders each containing 1 "atomix.conf" file)**

```
cd /path/to/your/onos/source/root
./tools/test/bin/atomix-gen-config 172.20.0.2 ~/atomix-1.conf
172.20.0.2 172.20.0.3 172.20.0.4
./tools/test/bin/atomix-gen-config 172.20.0.3 ~/atomix-2.conf
172.20.0.2 172.20.0.3 172.20.0.4
./tools/test/bin/atomix-gen-config 172.20.0.4 ~/atomix-3.conf
172.20.0.2 172.20.0.3 172.20.0.4
```

Ex: File "atomix-configs/atomix-1-conf/atomix.conf"...

```
{
    "cluster": {
        "clusterId": "onos",
        "node": {
            "id": "atomix-1",
            "address": "172.20.0.2:5679"
        },
        "discovery": {
            "type": "bootstrap",
            "nodes": [
                {
                    "id": "atomix-1",
                    "address": "172.20.0.2:5679"
                },
```

```
                {
                    "id": "atomix-2",
                    "address": "172.20.0.0.3:5679"
                },
                {
                    "id": "atomix-3",
                    "address": "172.20.0.4:5679"
                }
            ]
        }
    },
    "managementGroup": {
        "type": "raft",
        "partitions": 1,
        "partitionSize": 3,
        "members": [
            "atomix-1",
            "atomix-2",
            "atomix-3"
        ],
        "storage": {
            "level": "mapped"
        }
    },
    "partitionGroups": {
        "raft": {
            "type": "raft",
            "partitions": 3,
            "partitionSize": 3,
            "members": [
                "atomix-1",
                "atomix-2",
                "atomix-3"
            ],
            "storage": {
                "level": "mapped"
            }
        }
    }
}
```

a. If you have issues with python,install python3 and python3-pip

```
sudo apt update
sudo apt install -y python3 python3-pip /home/ubuntu/atomix-configs
```

b. Create a symlink to make "python" point to "python3"

```
sudo apt install -y python-is-python3
```

6. **Generate ONOS configuration files (don't need to do this if you already have a "onos-configs" folder containing 3 "cluster-[num].json" files)**

```
./tools/test/bin/onos-gen-config 172.20.0.5
~/onos-configs/cluster-1.json -n 172.20.0.2,172.20.0.3,172.20.0.4
```

```
./tools/test/bin/onos-gen-config 172.20.0.6
~/onos-configs/cluster-2.json -n 172.20.0.2,172.20.0.3,172.20.0.4
```

```
./tools/test/bin/onos-gen-config 172.20.0.7
~/onos-configs/cluster-3.json -n 172.20.0.2,172.20.0.3,172.20.0.4
```

Ex: File “onos-configs/cluster-1.json”...

```
{

        "name": "onos",
        "node": {
                "id": "172.20.0.5",
                "ip": "172.20.0.5",
                "port": 9876
        },

        "storage": [
                {
                        "id": "atomix-1",
                        "ip": "172.20.0.2",
                        "port": 5679
                },
                {
                        "id": "atomix-2",
                        "ip": "172.20.0.3",
                        "port": 5679
                },
                {
                        "id": "atomix-3",
                        "ip": "172.20.0.4",
                        "port": 5679
                }
        ]
}
```

7. **Continue to below section**

Restart setup

(once software downloaded & files all created, just repeat these steps)

1. **Stop & Remove all Docker containers & the created network (if any)**

```
docker stop $(docker ps -a -q)
docker rm $(docker ps -a -q)
docker network rm onos-atomix-net
docker system prune -a --volumes
```

2. **Pull in Atomix container**

```
docker pull atomix/atomix:3.1.5
```

3. **Create Docker network**

```
docker network create --subnet=172.20.0.0/16 onos-atomix-net
```

4. **Create 3 Docker containers for Atomix**

```
docker run -t -d --name atomix-1 --net onos-atomix-net --ip
172.20.0.2 -v
/home/ubuntu/atomix-configs/atomix-1-conf:/etc/atomix/conf
atomix/atomix:3.1.5 --config /etc/atomix/conf/atomix.conf
--ignore-resource

docker run -t -d --name atomix-2 --net onos-atomix-net --ip
172.20.0.3 -v
/home/ubuntu/atomix-configs/atomix-2-conf:/etc/atomix/conf
atomix/atomix:3.1.5 --config /etc/atomix/conf/atomix.conf
--ignore-resource

docker run -t -d --name atomix-3 --net onos-atomix-net --ip
172.20.0.4 -v
/home/ubuntu/atomix-configs/atomix-3-conf:/etc/atomix/conf
atomix/atomix:3.1.5 --config /etc/atomix/conf/atomix.conf
--ignore-resource
```

5. **Export environment vars for Atomix container IPs**

```
export OC1=172.20.0.2
export OC2=172.20.0.3
export OC3=172.20.0.4
```

6. **Create a Docker container for each of the 3 ONOS controllers, pre-import needed applications using ONOS_APPS="..."**

```
sudo docker run -t -d --name onos-1 --hostname onos-1 --net
onos-atomix-net --ip 172.20.0.5 -p 8181:8181 -e
ONOS_APPS="drivers,openflow,openflow-base,ofagent,netcfghostprovider,
lldpprovider,gui2,fwd,mfwd" onosproject/onos:2.2.1 >/dev/null
```

```
sudo docker run -t -d --name onos-2 --hostname onos-2 --net
onos-atomix-net --ip 172.20.0.6 -p 8182:8181 -e
ONOS_APPS="drivers,openflow,openflow-base,ofagent,netcfghostprovider,
lldpprovider,gui2,fwd,mfwd" onosproject/onos:2.2.1 >/dev/null
```

```
sudo docker run -t -d --name onos-3 --hostname onos-3 --net
onos-atomix-net --ip 172.20.0.7 -p 8183:8181 -e
ONOS_APPS="drivers,openflow,openflow-base,ofagent,netcfghostprovider,
lldpprovider,gui2,fwd,mfwd" onosproject/onos:2.2.1 >/dev/null
```

7. **Create new folders in ONOS containers**

```
docker exec onos-1 mkdir /root/onos/config
docker exec onos-2 mkdir /root/onos/config
docker exec onos-3 mkdir /root/onos/config
```

8. **Copy config files into new folder in ONOS containers**

```
docker cp ~/onos-configs/cluster-1.json
onos-1:/root/onos/config/cluster.json
docker cp ~/onos-configs/cluster-2.json
onos-2:/root/onos/config/cluster.json
docker cp ~/onos-configs/cluster-3.json
onos-3:/root/onos/config/cluster.json
```

9. **Restart ONOS containers to apply configurations**

```
docker restart onos-1 onos-2 onos-3
```

10. Go to your internet browser, try entering:

http://[IP_address]:8181/onos/ui/

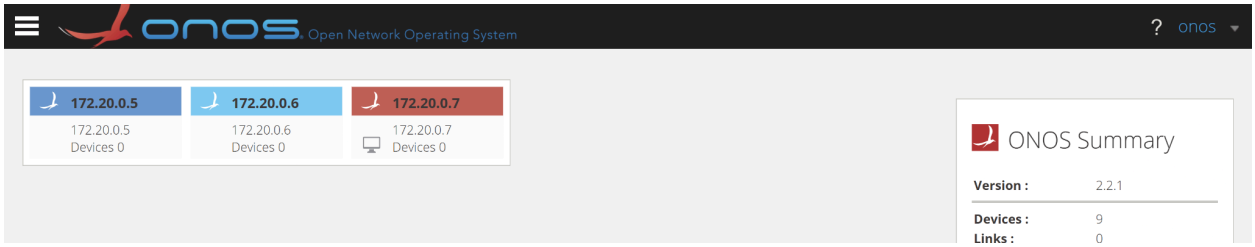
   a. If it doesn't work, try the below URLs:
   b. http://[IP_address]:8181
   c. http://[IP_address]:8182/onos/ui/
   d. http://[IP_address]:8182
   e. http://[IP_address]:8183/onos/ui/
   f. http://[IP_address]:8183

11. **When you see login page, enter default username & password**

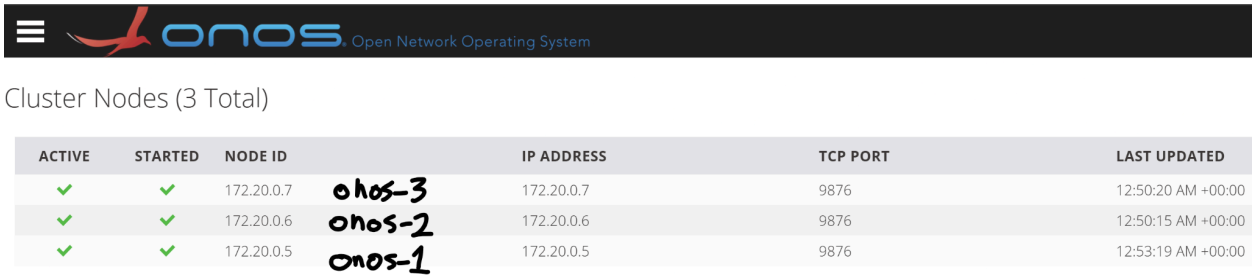
Username is *onos*

Password is *rocks*

12. **You should be taken to this topology page**



13. **In web UI, go to sidebar, "Cluster Nodes". Ensure you see green checks on all 3 ONOS Cluster nodes.**



a. **If any of the cluster nodes have a red-x on them, go back to terminal and restart that ONOS node. May need to RESTART and WAIT 5 min**

```
docker restart onos-[1,2, or 3]
```

14. **Clean Mininet**

```
sudo mn -c
```

15. **Run any custom Mininet topology Python file**

```
sudo python3 custom_topology.py
```

16. **In Mininet terminal, test connectivity of all hosts & switches**

mininet> `pingall`

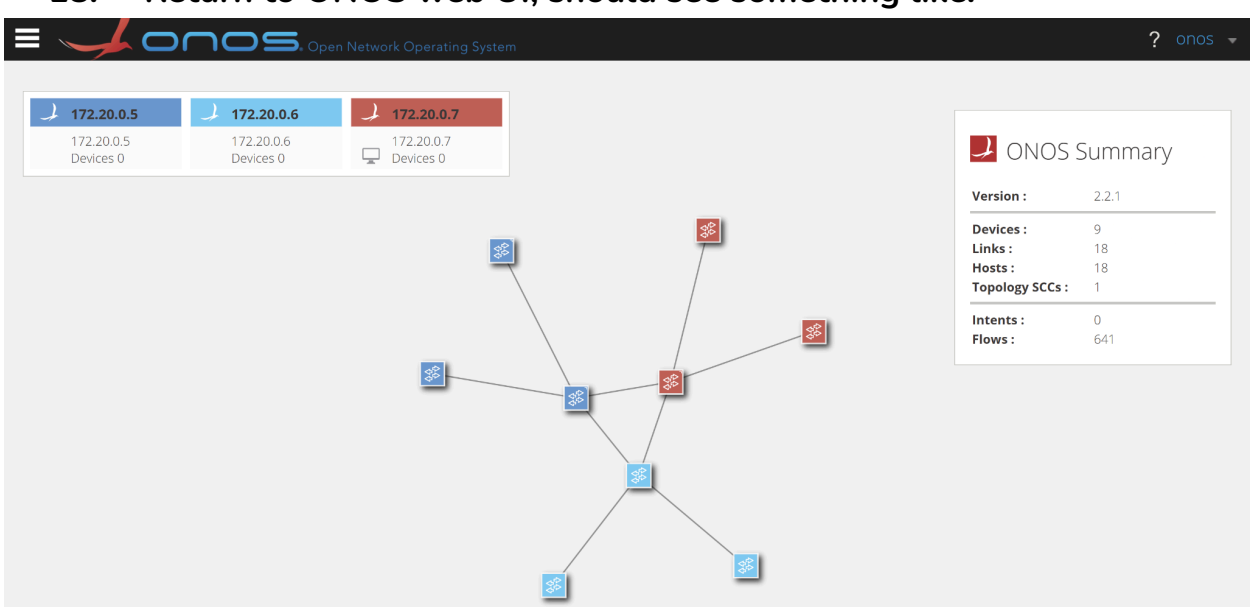
17. **If pingall fails, cleanup & try again:**

mininet> `exit`

```
sudo mn -c
sudo python3 custom_topology.py
```

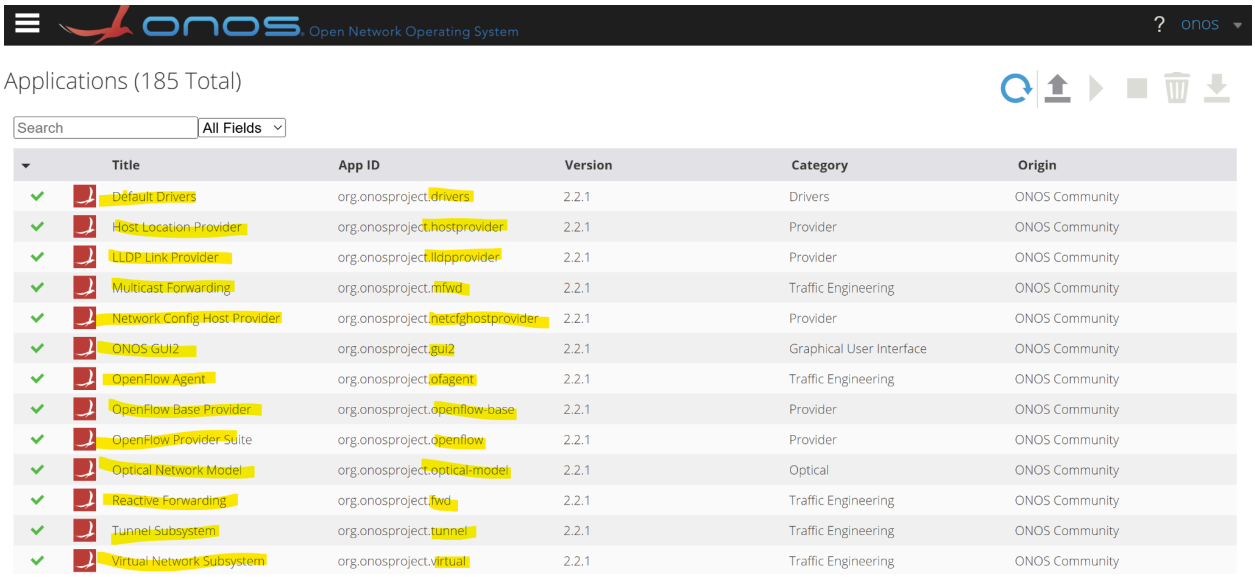
mininet> `pingall`

18. **Return to ONOS web UI, should see something like:**



Troubleshooting

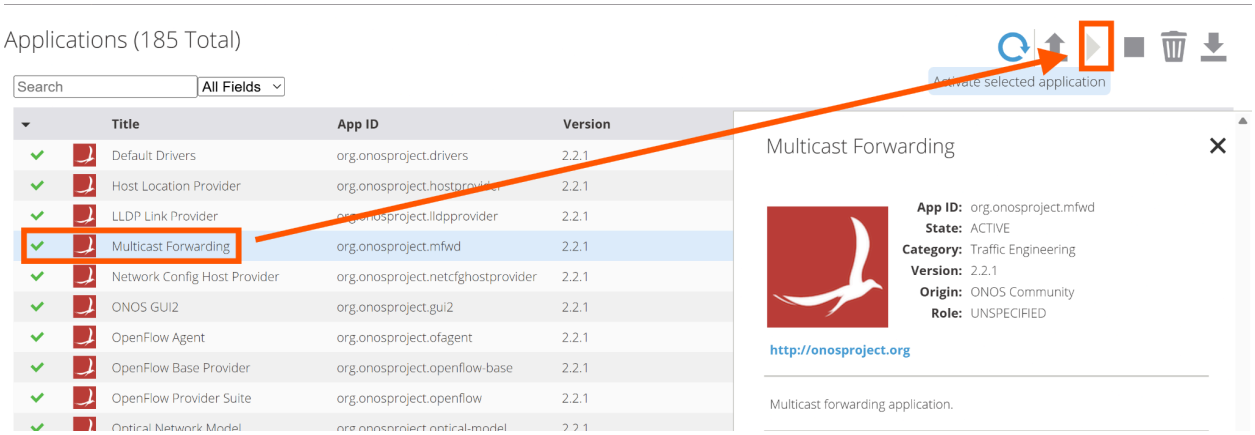
In web UI, ensure these are installed by going to sidebar, and clicking "Applications"



Title	App ID	Version	Category	Origin
Default Drivers	org.onosproject.drivers	2.2.1	Drivers	ONOS Community
Host Location Provider	org.onosproject.hostprovider	2.2.1	Provider	ONOS Community
LLDP Link Provider	org.onosproject.lldpprovider	2.2.1	Provider	ONOS Community
Multicast Forwarding	org.onosproject.mfwd	2.2.1	Traffic Engineering	ONOS Community
Network Config Host Provider	org.onosproject.netcfghostprovider	2.2.1	Provider	ONOS Community
ONOS GUI2	org.onosproject.gui2	2.2.1	Graphical User Interface	ONOS Community
OpenFlow Agent	org.onosproject.ofagent	2.2.1	Traffic Engineering	ONOS Community
OpenFlow Base Provider	org.onosproject.openflow-base	2.2.1	Provider	ONOS Community
OpenFlow Provider Suite	org.onosproject.openflow	2.2.1	Provider	ONOS Community
Optical Network Model	org.onosproject.optical-model	2.2.1	Optical	ONOS Community
Reactive Forwarding	org.onosproject.fwd	2.2.1	Traffic Engineering	ONOS Community
Tunnel Subsystem	org.onosproject.tunnel	2.2.1	Traffic Engineering	ONOS Community
Virtual Network Subsystem	org.onosproject.virtual	2.2.1	Traffic Engineering	ONOS Community

If not, click on application(s) and import it(them) by pressing triangle button in the opened window.

Ex:



THIS DOES NOT UPDATE ACTUAL SETUP, IT ONLY IMPORTS APPLICATION!!! To update actual setup, return to terminal:

```
docker restart onos-1 onos-2 onos-3
```